

VirtualWorks Lab Internship

Task 1 — Adverse Drug Reaction (ADR) Case Analysis

Five simulated case scenarios

About this submission

This assignment presents five fictional ADR cases and shows how each case can be reviewed in a structured way. The focus is on the patient and reaction details, suspected medicine, other exposures, causality factors, management and outcome.

Disclaimer

All cases in this report are simulated for internship training. Patient initials, clinical events, dates, manufacturers and batch information are fictional and should not be treated as real pharmacovigilance reports.

Causality approach

The WHO-UMC approach was used as the main framework. Each case considers the timing of the reaction, whether the event is consistent with the known effect of the medicine, dose-response where useful, dechallenge and rechallenge. Rechallenge was not suggested when it would be inappropriate or unsafe.

ADR-001 — Pharmacological / predictable

Patient profile	AB 58 years Female 72 kg
Suspected ADR	Bilateral ankle swelling (peripheral edema)
Onset	Day 10 after starting treatment
Clinical description	The patient developed swelling around both ankles. There was no shortness of breath or other sign suggesting acute heart failure.
Suspected drug	Amlodipine
Dose / route / frequency	10 mg / Oral / Once daily
Indication	Hypertension
Concomitant medicines	Atorvastatin 20 mg once daily
OTC / supplements	None reported
Outcome	Swelling gradually disappeared within 7 days after amlodipine was stopped.

Causality assessment

Temporal relationship: The swelling started 10 days after amlodipine was started.

Plausibility: Peripheral edema is a known adverse effect of amlodipine and can be more common at higher doses.

Dose-response: A clear dose-response could not be established because there was no documented previous dose reduction.

Dechallenge: Positive. The swelling resolved after the suspected drug was withdrawn.

Rechallenge: Not done. Rechallenge was not considered necessary for a predictable, non-serious reaction.

Final WHO-UMC assessment: Probable / Likely

Why this assessment: The timing was suitable, the reaction is a known effect of amlodipine, and the symptoms improved after withdrawal.

Management

Amlodipine was stopped. Blood pressure was monitored and another antihypertensive could be considered if required.

ADR-002 — Pathological / toxic

Patient profile	CD 46 years Male 81 kg
Suspected ADR	Raised liver enzymes with fatigue and nausea
Onset	Week 5 after starting treatment
Clinical description	The patient reported mild nausea and fatigue. Blood tests showed an increase in ALT and AST compared with baseline.
Suspected drug	Methotrexate
Dose / route / frequency	15 mg / Oral / Once weekly
Indication	Rheumatoid arthritis
Concomitant medicines	Folic acid 5 mg once weekly on a separate day
OTC / supplements	Occasional paracetamol; no herbal products reported
Outcome	Liver enzyme values gradually improved after methotrexate was withdrawn.

Causality assessment

Temporal relationship: The liver enzyme changes appeared after several weeks of methotrexate treatment.

Plausibility: Methotrexate is known to cause liver toxicity, particularly with continued exposure.

Dose-response: A dose-response relationship could not be assessed from the available simulated information.

Dechallenge: Positive. Liver enzyme values improved after stopping methotrexate.

Rechallenge: Not done because restarting the suspected drug could expose the patient to further liver injury.

Final WHO-UMC assessment: Probable / Likely

Why this assessment: The reaction occurred after a reasonable exposure period, is consistent with a known methotrexate toxicity, and improved after withdrawal.

Management

Methotrexate was withheld and liver function was followed. An alternative treatment was considered for the underlying condition.

ADR-003 — Allergic / hypersensitivity

Patient profile	EF 29 years Male 68 kg
Suspected ADR	Generalized urticaria with itching
Onset	About 2 hours after the first dose
Clinical description	The patient developed widespread raised, itchy wheals. No breathing difficulty, hypotension or airway swelling was reported.
Suspected drug	Amoxicillin
Dose / route / frequency	500 mg / Oral / Three times daily
Indication	Bacterial respiratory infection
Concomitant medicines	Paracetamol 500 mg as needed
OTC / supplements	None reported
Outcome	The rash and itching settled within 24 hours after amoxicillin was stopped and treatment was given.

Causality assessment

Temporal relationship: The reaction started approximately two hours after the first dose.

Plausibility: Urticaria appearing soon after a beta-lactam antibiotic is compatible with an immediate hypersensitivity reaction.

Dose-response: Not assessed because this was the first dose.

Dechallenge: Positive. The symptoms settled after the drug was discontinued and the reaction was treated.

Rechallenge: Not done. Re-exposure was not considered appropriate after a suspected allergic reaction.

Final WHO-UMC assessment: Probable / Likely

Why this assessment: The very short time between exposure and reaction, together with the typical presentation, supports a probable relationship.

Management

Amoxicillin was stopped. An antihistamine was given and the patient was observed for worsening symptoms.

ADR-004 — Pharmacogenetic / idiosyncratic

Patient profile	GH 35 years Female 61 kg
Suspected ADR	Severe skin reaction consistent with SJS/TEN spectrum
Onset	Day 18 after starting treatment
Clinical description	The patient developed fever followed by painful red skin lesions and involvement of mucosal surfaces.
Suspected drug	Carbamazepine
Dose / route / frequency	200 mg / Oral / Twice daily
Indication	Focal seizures
Concomitant medicines	None reported
OTC / supplements	None reported
Outcome	The patient recovered after carbamazepine was stopped and specialist inpatient treatment was provided.

Causality assessment

Temporal relationship: The reaction began around 18 days after treatment was started, which is compatible with a delayed serious skin reaction.

Plausibility: Carbamazepine is associated with serious cutaneous reactions, and genetic factors can increase susceptibility in some patients.

Dose-response: Not established. A dose-response relationship is not expected to be necessary for an idiosyncratic immune-mediated reaction.

Dechallenge: Positive in this simulated case. The condition improved after withdrawal and treatment.

Rechallenge: Not done and would be inappropriate because of the seriousness of the suspected reaction.

Final WHO-UMC assessment: Probable / Likely

Why this assessment: The timing and clinical pattern fit a known serious reaction to carbamazepine. Genetic susceptibility is included as a possible mechanism, not as a confirmed test result.

Management

Carbamazepine was stopped immediately and the patient received urgent specialist care. Permanent avoidance was documented.

ADR-005 — Teratogenic / developmental toxicity

Patient profile	IJ 31 years Female 64 kg
Suspected ADR	Fetal neural-tube defect identified during prenatal assessment
Onset	Detected during pregnancy following early-pregnancy exposure
Clinical description	In this simulated case, valproate treatment continued into early pregnancy and prenatal assessment identified a neural-tube defect.
Suspected drug	Sodium valproate / valproic acid
Dose / route / frequency	500 mg / Oral / Twice daily
Indication	Epilepsy
Concomitant medicines	Folic acid supplementation reported
OTC / supplements	None reported
Outcome	The pregnancy was followed by a specialist multidisciplinary team and the outcome was documented for pharmacovigilance assessment.

Causality assessment

Temporal relationship: The exposure occurred during the early developmental period when neural-tube formation takes place.

Plausibility: Valproate exposure during pregnancy is a recognized risk factor for congenital malformations, including neural-tube defects.

Dose-response: Not assessed in this simulated case.

Dechallenge: Not applicable to the already established developmental outcome.

Rechallenge: Not applicable and not appropriate.

Final WHO-UMC assessment: Probable / Likely

Why this assessment: The timing of exposure and the type of fetal outcome are compatible with the known developmental risks associated with valproate. This is a simulated assessment and does not prove causation in an individual pregnancy.

Management

Specialist obstetric, fetal-medicine and neurology follow-up was arranged, with review of the maternal treatment plan.

Key points from the cases

1. A reaction occurring after a medicine is started should be checked against the expected timing of the suspected ADR.
2. A known pharmacological effect or a recognizable clinical pattern can strengthen the assessment.
3. Improvement after withdrawal (positive dechallenge) is useful supporting evidence, although it is not proof by itself.
4. Rechallenge is not required for every case and should not be recommended when it may put a patient at risk.
5. Serious reactions need prompt clinical management and appropriate documentation.
6. In a training exercise, simulated information should always be clearly separated from real patient safety data.

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